King Fahd University of Petroleum and Minerals
Department of Mathematics
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Stat 201 Major Exam I Term 222

Original exam date: 1 March 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes Original course: STAT 319

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1. Heart failure is due to either natural occurrences (87%) or foreign objects (13%). Outside factors are related to induced substances or foreign objects. Natural Occurrences are caused by arterial blockage, disease, and infection. Assume that causes of heart failure for the individuals are independent. What is the probability that the fourth patient with heart failure who enters the emergency room is the first one due to nature occurrences?

(a) 0.8700

(b) 0.2568

(c) 0.0019

(d) 0.1300

(e) 0.6126

2. A manufacturer produces electrical components of which 3% are faulty. The components are sold in boxes of 20 and the boxes are sold to wholesalers in batches of 10. A box is rejected if there is more than one faulty component in it and a batch is rejected if more than one box in the batch is rejected. What is the probability that a batch is rejected?

(a) 0.657

(b) 0.120

(c) 0.880

(d) 0.341

(e) 0.242

3. An inspector working for a manufacturing company has a 98% chance of correctly identifying defective items and a 0.5% chance of incorrectly classifying a good item as defective. The company has evidence that 1% of the items its line produces are nonconforming. If an item selected at random is classified as non defective, what is the probability that it is indeed good?

(a) 0.0041

(b) 0.9996

(c) 0.0141

(d) 0.9821

(e) 0.9052

4. Hits to a high-volume Web site are assumed to follow a Poisson distribution with a mean of 10,000 per day. Approximate the probability of more than 20,000 hits in a day.

(a) 0.00

(b) 10.00

(c) ∞

(d) 0.50

(e) 1.00

5. Strands of copper wire from a manufacturer are analyzed for strength and conductivity. The results from 100 strands are as follows:

	Strength	
	High	Low
High conductivity	74	8
Low conductivity	15	3

If a strand is randomly selected, what is the probability that its conductivity is low or its strength is low?

(a) 0.89

(b) 0.11

(c) 0.74

(d) 0.26

(e) 0.27

6. The life of automobile voltage regulators has an exponential distribution with a mean life of five years. You purchase a six-year-old automobile, with a working voltage regulator and plan to own it for five years. What is the probability that the voltage regulator fails during your ownership?

(a) 0.3679

(b) 0.8892

(c) 0.1108

(d) 0.6321

(e) 0.5717

7. An e-mail message will arrive at a time uniformly distributed between 9:00 a.m. and 11:00 a.m. You check e-mail at 9:15 a.m. and every 30 minutes afterward. What is the probability that the message arrives more than 15 minutes before you view it?

(a) 0.33

(b) 0.99

(c) 0.25

(d) 0.50

(e) 0.75

8. The lifetime (in hours) of a semiconductor laser has a lognormal distribution with $\theta = 4$ and $\omega = 2$. What is the probability that the lifetime exceeds its mean?

(a) 0.69146

(b) 0.15866

(c) 0.30854

(d) 0.50000

(e) 0.84134

9. Errors in an experimental transmission channel are found when the transmission is checked by a certifier that detects missing pulses. The number of errors found in an eight-bit byte is a random variable with the following distribution:

$$F(x) = \begin{cases} 0, & x < 1 \\ 0.7, & 1 \leq x < 4 \\ 0.9, & 4 \leq x < 7 \\ 1, & 7 \leq x \end{cases}$$

$P(X \leq 2)$ is equal to

(a) 0.85

(b) 0.10

(c) 0.90

(d) 0.70

(e) 0.30

10. The time that it takes a randomly selected sports car from the same brand to accelerate from 0 to 60 miles per hour (mph) is normally distributed with mean 6.2 seconds and standard deviation 0.15 seconds. Suppose 6 cars are selected at random, and let $X_1, \cdots, X_6$ denote their times to reach 60 mph. What is the probability that the total time $T = \sum_{i=1}^{6} X_i$ is between 36.5 and 37.5 seconds?

(a) 0.9648

(b) 0.7658

(c) 0.6658

(d) 0.0281

(e) 0.1658

11. The operational lifetime X, in years, of a battery powered watch has probability density function

$$f(x) = \begin{cases} c\,x(6-x), & 3 \le x \le 6 \\ 0, & \text{otherwise.} \end{cases}$$

Find the probability that the watch has an operational lifetime in excess of 4 years.

(a) 0.534

(b) 0.286

(c) 0.318

(d) 0.519

(e) 0.056

12. Suppose that the lifetime of a component (in hours) modeled with a Weibull distribution with parameters $\beta = 0.5$ and $\delta = 4000$ hours. The median lifetime is equal to

(a) 3330.22 hours

(b) 1921.81 hours

(c) 3843.62 hours

(d) 832.555 hours

(e) 480.453 hours

13. The number of cracks in a section of interstate highway that are significant enough to require repair is assumed to follow a Poisson distribution with a mean of two cracks per mile. What is the probability that at least one crack requires repair in $\frac{1}{2}$ mile of highway?

(a) 0.7397

(b) 0.6321

(c) 0.6572

(d) 0.3428

(e) 0.3679

14. An e-mail filter is planned to separate valid e-mails from spam. The word free occurs in 60% of the spam messages and only 4% of the valid messages. Also, 20% of the messages are spam. Determine the probability that a message is valid given that it does not contain free.

(a) 0.9600

(b) 0.0339

(c) 0.9999

(d) 0.9057

(e) 0.3600

15. A research study uses 900 men under the age of 55. Suppose that 30% carry a marker on the male chromosome that indicates an increased risk for high blood pressure. If 10 men are selected randomly and tested for the marker, what is the probability that more than 1 has the marker?

(a) 0.5433

(b) 0.4567

(c) 0.0276

(d) 0.1478

(e) 0.8522

16. A batch of parts contains 100 from a local supplier of tubing and 200 from a supplier of tubing in the next state. If four parts are selected randomly and without replacement, what is the probability that two or more parts in the sample are from the local supplier?

(a) 0.890

(b) 0.592

(c) 0.110

(d) 0.408

(e) 0.396

17. Let A and B be two events such that $P(A/B) = 2P(A/\bar{B})$ and $P(B) = 0.32$ then $P(B/A)$ is

(a) 0.52

(b) 0.48

(c) 0.68

(d) 0.50

(e) 0.00

18. Let R have probability mass function (pmf) $p(r) = \frac{1}{8}$ for $r = 1, 2, \cdots, 8$. Find the variance of R, $\text{var}(R)$.

(a) 3.70

(b) 4.50

(c) 10.50

(d) 2.67

(e) 5.25

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	Geometric: $P(X = 4) = 0.13^3(0.87) = 0.0019$.
2	**(d)**	Box: $P = 0.1198$; batch: $P(Y > 1)$ with $Y \sim \text{Bin}(10, 0.1198)$ is 0.341.
3	**(b)**	$\frac{0.99(0.995)}{0.99(0.995)+0.01(0.02)} \approx 0.9998$; closest choice 0.9996.
4	**(a)**	$X \approx N(10^4, 10^4)$: $P(Z > 100) \approx 0.00$.
5	**(d)**	$P = \frac{8+15+3}{100} = 0.26$.
6	**(d)**	Memoryless: $P(X < 5) = 1 - e^{-1} = 0.6321$.
7	**(d)**	Favourable time $4 \times 15 = 60$ of 120 minutes: 0.50.
8	**(b)**	$E(X) = e^6$: $P(Z > 1) = 0.15866$.
9	**(d)**	$P(X \leq 2) = F(2) = 0.70$.
10	**(b)**	$T \sim N(37.2, 0.135)$: $\Phi(0.82) - \Phi(-1.91) = 0.7658$.
11	**(d)**	$c = \frac{1}{18}$; $P(X > 4) = \frac{1}{18} \cdot \frac{28}{3} = 0.519$.
12	**(b)**	$x = \delta(\ln 2)^{1/\beta} = 4000(\ln 2)^2 = 1921.81$.
13	**(b)**	$\lambda = 1$: $P(X \geq 1) = 1 - e^{-1} = 0.6321$.
14	**(d)**	$\frac{0.96(0.8)}{0.4(0.2)+0.96(0.8)} = 0.9057$.
15	**(e)**	Hypergeometric $N = 900, K = 270, n = 10$: $P(X > 1) = 0.8522$.
16	**(d)**	Hypergeometric $N = 300, K = 100, n = 4$: $P(X \geq 2) = 0.407 \approx 0.408$.
17	**(b)**	$P(A) = 1.32a$, $P(B \mid A) = \frac{0.64}{1.32} = 0.48$.
18	**(e)**	$E(R) = 4.5$, $E(R^2) = 25.5$, var $= 5.25$.